

Access Control System

Cryptographic Authentication & Role-Based Authorization

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What is Access Control?

Access Control restricts who can view or use resources in a computing environment. Cryptography makes this process secure.

Authentication

Verify user identity before granting access

Authorization

Determine what an authenticated user may do

Encryption

Protect passwords using cryptographic algorithms

Analogy: A hospital where doctors, nurses, and visitors each have different levels of access — enforced by cryptography.

How Cryptography Secures Access

01

Password Hashing

Passwords are transformed using SHA-256 into a fixed hash that cannot be reversed.

02

Salt

A unique random value added to each password before hashing prevents duplicate hashes.

03

PBKDF2

The hashing runs 100,000 iterations, making brute-force attacks extremely slow.

04

RBAC

Role-Based Access Control assigns each user a role that defines which resources they can access.

Login Page — Authentication

Access Control System

Cryptographic authentication & role-based authorization

USERNAME

wafaa

PASSWORD

AUTHENTICATE +

- click an account to auto-fill -

wafaa

Wafaa#2925

ADMIN

dowa

Dowa#2925

USER

tasneem

Tasneem#2925

USER

thekra

Thekra#789

GUEST

Secure Login Form

Username and password fields with cryptographic verification

4 User Accounts

Wafaa (Admin), Dowa & Tasneem (User), Thekra (Guest)

Password Hashing

Passwords converted to SHA-256 hash — never stored as plain text

Role Assignment

After login, the system assigns the user a role determining their access

Wafaa — Admin Dashboard

SecureACS

wafaaADMINSIGN OUT

SESSION OVERVIEW

5
GRANTED

0
DENIED

5
TOTAL REQUESTS

RESOURCES — CLICK TO REQUEST ACCESS

**View Reports**
Read analytics & data

ALLOWED

**Update Profile**
Edit personal settings

ALLOWED

**View Logs**
Audit trail access

ALLOWED

**Manage Users**
Add/remove accounts

ALLOWED

**Manage Keys**
Cryptographic key store

ALLOWED

**Delete Records**
Permanent data removal

ALLOWED

CRYPTOGRAPHIC DETAILS

ALGORITHM
PBKDF2-HMAC-SHA256

SALT (HEX)
f2a9c1b8d3e04712fa56

STORED HASH (FIRST 32 CHARS)
9c3d2e1b8f074a6c2d9e3f1a8b7c04a1_

ITERATIONS
100,000 rounds

ACCESS LOG

☒

ACCESS GRANTED
wafaa - View Logs
12:46:42 م

☒

ACCESS GRANTED
wafaa - Manage Users
12:46:41 م

All 6 Resources: ALLOWED

Full system access granted

Admin Permissions

- ✓ View Reports
- ✓ Update Profile
- ✓ View Logs
- ✓ Manage Users
- ✓ Manage Keys
- ✓ Delete Records

Dowa — User Dashboard

SecureACS

dowa

USER

SIGN OUT

SESSION OVERVIEW

2
GRANTED

0
DENIED

2
TOTAL REQUESTS

RESOURCES — CLICK TO REQUEST ACCESS



View Reports
Read analytics & data

ALLOWED



Update Profile
Edit personal settings

ALLOWED



View Logs
Audit trail access

LOCKED



Manage Users
Add/remove accounts

LOCKED



Manage Keys
Cryptographic key store

LOCKED



Delete Records
Permanent data removal

LOCKED

CRYPTOGRAPHIC DETAILS

ALGORITHM

PBKDF2-HMAC-SHA256

SALT (HEX)

a1e2d3b0f4c5883490bc

STORED HASH (FIRST 32 CHARS)

3f1d8e2a0c9b5d4f7e2a1c3b8d0f9e3b_

ITERATIONS

100,000 rounds

ACCESS LOG

☒

ACCESS GRANTED

dowa → Update Profile

12:49:01

☒

ACCESS GRANTED

dowa → View Reports

12:49:01

2 of 6 Resources: ALLOWED

Limited access only

User Permissions

✓ View Reports

✓ Update Profile

✗ View Logs

✗ Manage Users

✗ Manage Keys

✗ Delete Records

GUEST

Thekra — Guest Dashboard

SecureACS

T thekra GUEST SIGN OUT

SESSION OVERVIEW

3 GRANTED

0 DENIED

3 TOTAL REQUESTS

RESOURCES — CLICK TO REQUEST ACCESS

View Reports
Read analytics & data

ALLOWED

Update Profile
Edit personal settings

LOCKED

View Logs
Audit trail access

LOCKED

Manage Users
Add/remove accounts

LOCKED

Manage Keys
Cryptographic key store

LOCKED

Delete Records
Permanent data removal

LOCKED

Access granted: View Reports

CRYPTOGRAPHIC DETAILS

ALGORITHM
PBKDF2-HMAC-SHA256

SALT (HEX)
c9d4e3b2a1f076238e1a

STORED HASH (FIRST 32 CHARS)
8e2a4f1d0c3b9e5a7f2d1a4c8b3e0f1d_

ITERATIONS
100,000 rounds

ACCESS LOG

✓ ACCESS GRANTED
thekra → View Reports
12:49:43 صباحاً

✓ ACCESS GRANTED
thekra → View Reports
12:49:37 صباحاً

ACCESS GRANTED

1 of 6 Resources: ALLOWED

Minimum access only

Guest Permissions

✓ View Reports

✗ Update Profile

✗ View Logs

✗ Manage Users

✗ Manage Keys

✗ Delete Records

Role Permissions Summary

Resource	Admin (Wafaa)	User (Dowa/Tasneem)	Guest (Thekra)
View Reports	✓	✓	✓
Update Profile	✓	✓	✗
View Logs	✓	✗	✗
Manage Users	✓	✗	✗
Manage Keys	✓	✗	✗
Delete Records	✓	✗	✗

How the Code Works



User enters
password



System adds
unique salt



PBKDF2-SHA256
hashes it



Hash compared
to stored hash



Role assigned
& access granted

hash_password()

```
salt = os.urandom(16)
key = pbkdf2_hmac(
    "sha256", password,
    salt, 100000)
```

verify_password()

```
_, attempt = hash_password(
    input, stored_salt)
return attempt == stored
```

access_resource()

```
allowed = RESOURCES[role]
if resource in allowed:
    print("GRANTED")
else: print("DENIED")
```

Real-World Applications

Banking Systems

Tellers access accounts, managers approve transfers, all cryptographically secured.

Healthcare

Doctors access records, nurses view vitals, patients see only their own data.

Cloud Platforms

AWS, Azure & Google Cloud use role-based policies for resource management.

Government

Classified documents use cryptographic multi-level access control.

Mobile Apps

OAuth and JWT are cryptographic standards used globally for app access.

Universities

Students, professors, and admins each have different access to university systems.

Key Takeaways

01

Access Control is the backbone of system security — only authorized users can reach sensitive resources.

02

Cryptography (SHA-256, PBKDF2, Salt) protects stored passwords from theft even if the database is compromised.

03

RBAC allows fine-grained permission management — each role accesses only what it needs.

04

Our SecureACS demo implements login, hashing, roles, access logs, and live enforcement in a real web interface.

Sources & References

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[4] Ferraiolo et al. (2001). Proposed NIST standard for role-based access control. ACM TISSEC, 4(3), 224-274.

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[6] Menezes, A. et al. (1996). Handbook of Applied Cryptography. CRC Press.